ToC

Date : 07/09/2022

IMPROVEMENT CIE

Scheme & Solution

1) $M = (Q, \Sigma, \Gamma, \delta, q_0, B, F)$

Q - Set of finite states

Σ - Set of i/p alphabets

Γ - Set of tape symbols

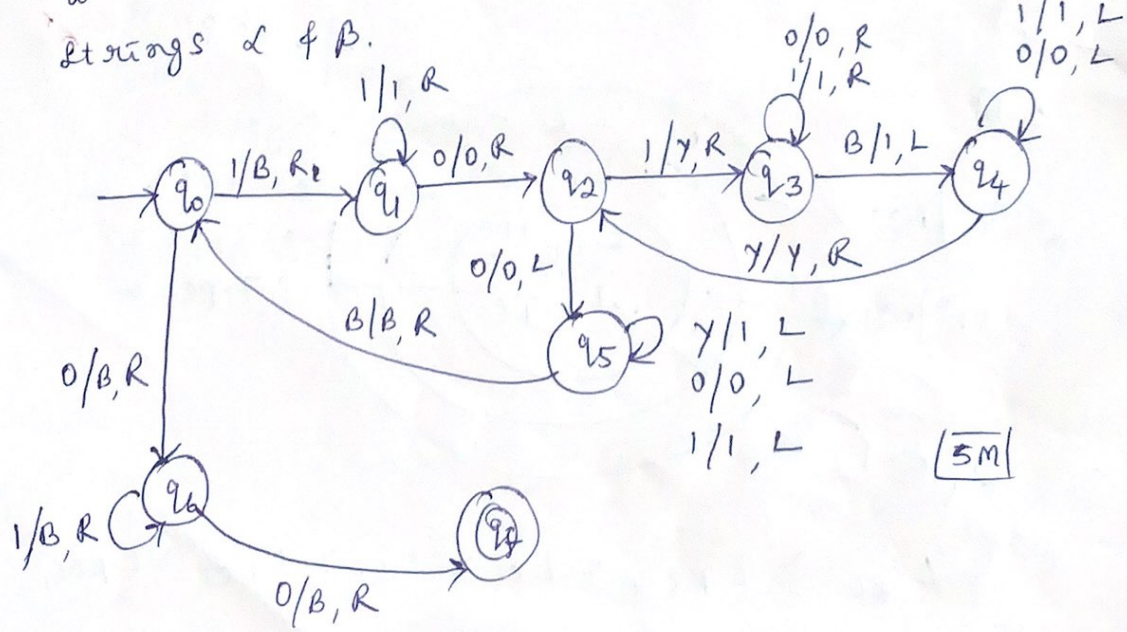
δ - $Q \times \Gamma$ to $Q \times \Gamma \times \{L, R\}$ [1M]

q_0 - start state

B - Blank character

The language of a TM: [1M]

Let $M = (Q, \Sigma, \Gamma, \delta, q_0, B, F)$ be a TM. Then $L(M)$ is the set of strings w in Σ^* such that $q_0 w \vdash^* \alpha p \beta$ for some state p in F & any tape strings α & β.



[5M]

2 ID tracing for $m = 5$, $n = 2$.

2.a (i) Non-deterministic TM

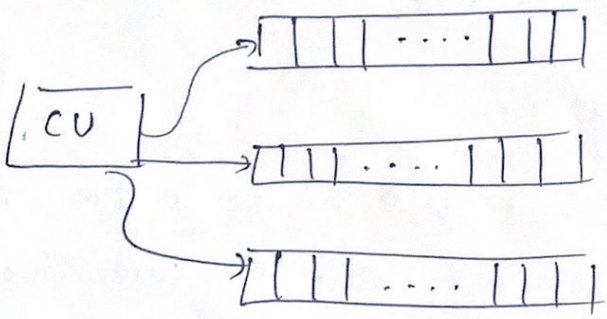
def^n, Transition fn: } 2M

$$\delta \rightarrow Q \times \Gamma \rightarrow 2^{Q \times \Gamma \times \{L, R\}}$$

For any i/p & current state → multiple choices.

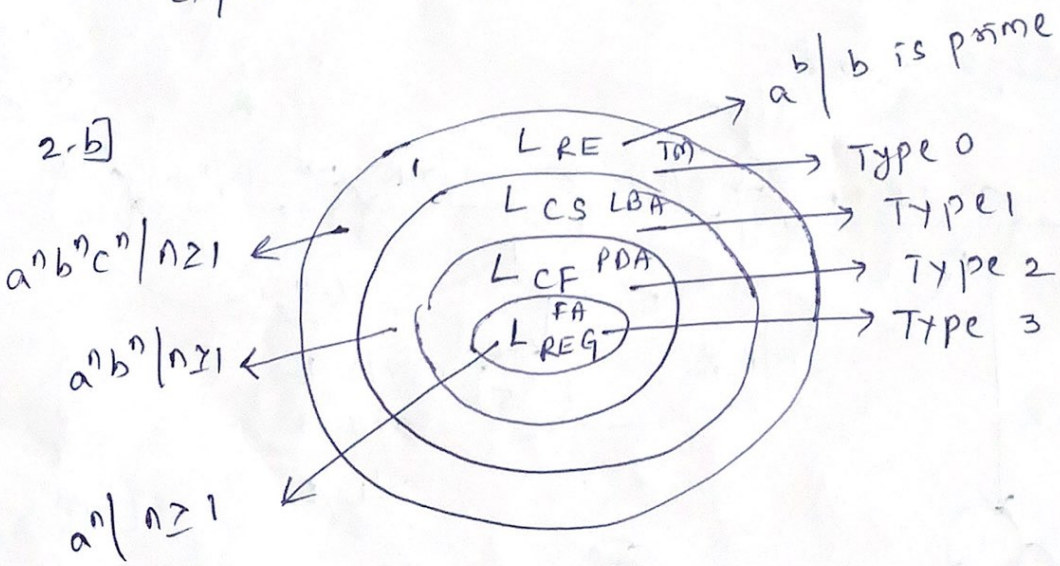
Diagram & detailed explanation

(ii) Multitape TM 3.5M



explanation & example.

2.b



$L_{RE} > L_{REG} > L_{CS} > L_{CF} > L_{DCF} > L_{REG}$

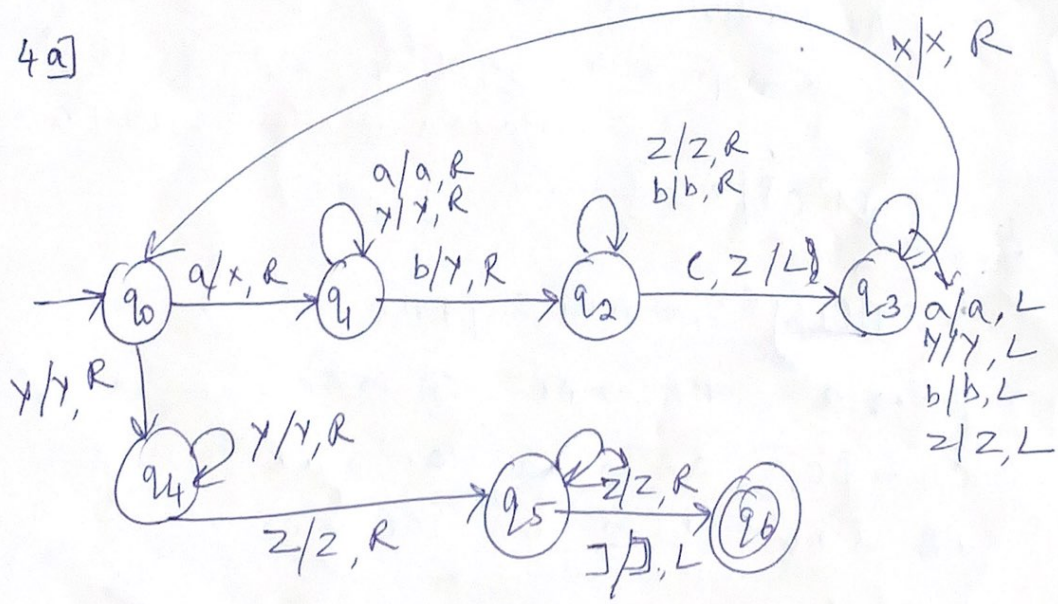
...ep of
...ith diagram → 3 * 2 = 6M

PCP - Given a set of dominos, we need to find a sequence of dominos such that the top & bottom strings are same.

* Each domino can be used any no of times & order doesn't matter.

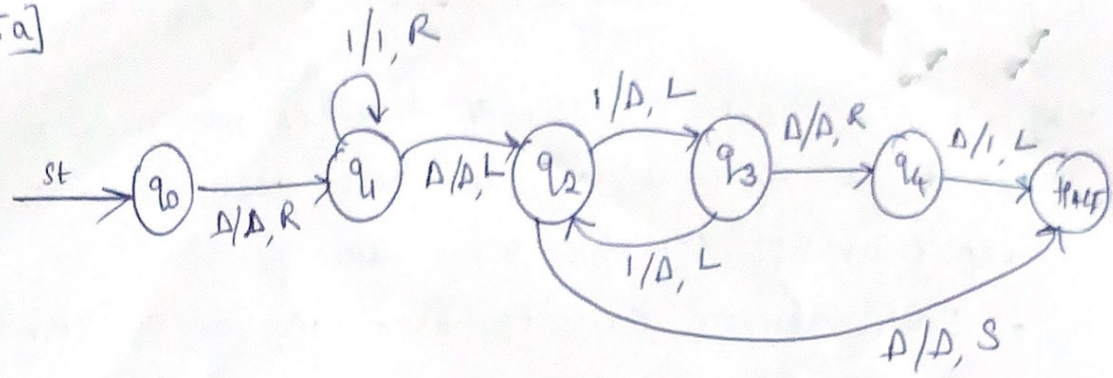
PCP with two lists X = (10, 011, 101) & Y = (101, 11, 011) has no solution. & PCP is an undecidable problem.

4a]



4b] halting problem description with an example - 3M

5a]



5b] Def^n of unrestricted grammar: 1M

All productions are of the form

$\alpha \to \beta$ with ~~$|\beta| \geq |\alpha|$~~, $\alpha, \beta \in \{V \cup T\}^*$

α contains a variable. i.e ~~$|LHS| \leq |RHS|$~~

$L = \{a^n b^n c^n \mid n \geq 1\}$ → 3M

$S \to ABCS_1 \mid ABC$ $\quad S_1 \to ABCS_1 \mid ABC$

$BA \to AB$, $CA \to AC$, $A \to a$

$CB \to BC$, $aA \to aa$, $aB \to ab$

$bB \to bb$, $bC \to bc$, $cC \to cc$

Derivation for the string aabbcc - 2M

~~$S \to FS_1$~~

~~$S \to \varepsilon ABCS_1$~~

~~$S \to aABCS_1$~~

~~$S \to aaBCS_1$~~

~~$S \to aabCS_1$~~

~~$S \to aa1$~~

$S \to ABCS_1$

$S \to ABCABC$

$ABACBC$

$AABCBC$

$AABBCC$

$aABBCC$

$aaBBCC$

$aabBCC$

$aabbCC$

aabbcC

aabbcc